Parking assistant

A distance reading changes a decision

Today’s robot challenge

A broad cardboard wall

A measured distance threshold

A record of why the robot stopped

GREEN steady: on

ORANGE flashing: charge

RED flashing: stop + mentor

Center button lights. A letter on the display is your program’s output.

Sensor input and if conditions

```
stop_mm = 150
gap = distance_sensor.distance(port.C)
if 0 <= gap <= stop_mm:
    await signal("P")
```

Our prediction

At 220, 150 and 90 mm, is the robot near a 150 mm limit?

What should happen with an invalid reading of -1?

Reading fragment from the complete supplied project

Our robot and build jobs

Driver edits and starts after “clear.”

Navigator predicts and resets the stopped robot.

Observer watches, measures and records.

Other team builds. Everyone rotates jobs.

Robot turns: rotate every 4 minutes. Retests: every 2 minutes.

Evidence from our test

We changed _____.

We predicted _____.

We observed _____.

Our next test would _____.

What we learned and noticed

What did you learn about sensors and decisions?

What did you observe as the robot neared the wall?

What would you change in the garage or next test?

Save your team project. Store your build. Reset the robot and parts.